

Paddy Crop Analysis Using Python

1. Project Overview

The **Paddy Crop Analysis Using Python** project aims to analyse a paddy cultivation dataset using Python to identify factors that influence crop yield. The project includes data cleaning, preprocessing, exploratory data analysis (EDA), statistical analysis, data visualization, and predictive suggestions. Using libraries such as **Pandas, NumPy, Matplotlib and Seaborn**, the project transforms raw agricultural data into meaningful insights to support better farming decisions and improve paddy productivity.

2. Data Source

- **Dataset Name:** Paddy Dataset
- **Source:** UCI Machine Learning Repository – Paddy Dataset (<https://archive.ics.uci.edu/dataset/1186/paddy+dataset>)
- **File Type:** CSV (.csv)
- **Number of Records:** 2,789
- **Number of Attributes:** 45

The dataset contains information related to paddy cultivation, including crop variety, soil type, seed rate, fertilizer usage, weather conditions, irrigation, pest management, and yield. It is used to analyse the factors affecting paddy production and generate meaningful insights using Python.

3. Problem Statement

Paddy yield is influenced by several factors, including soil type, seed rate, fertilizer application, weather conditions, irrigation, and pest management. Analysing these factors manually is difficult and time-consuming. This project aims to analyse the paddy dataset using Python to identify patterns, understand the factors affecting crop yield, and generate insights that support better agricultural planning and decision-making.

4. Objectives

- To consolidate and preprocess the Paddy Crop dataset by integrating key agricultural attributes such as paddy variety, soil type, seed rate, fertilizer application, irrigation practices, and crop yield for effective analysis.
 - To analyze the distribution and characteristics of major cultivation factors, including paddy varieties, soil types, land area, and farming practices, to understand overall dataset patterns.
 - To compare paddy yield variations across different paddy varieties and soil types to identify factors associated with higher and lower agricultural productivity.
 - To examine the relationship between cultivation factors such as Agriblock, paddy variety, soil type, nursery practices, and crop yield using statistical analysis and data visualization techniques.
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5. Attribute (Column / Feature) Details

Attribute Name	Description
Hectares	Total cultivated land area under paddy cultivation
Agriblock	Agricultural block/region where the field is located
Variety	Variety of paddy cultivated
Soil Types	Type of soil used for cultivation
Seedrate (Kg)	Quantity of seeds used for cultivation (in kilograms)
Nursery	Nursery preparation details
Nursery Area (Cents)	Area allocated for nursery cultivation
DAP_20days	Quantity of DAP fertilizer applied after 20 days
Urea	Quantity of urea fertilizer applied
Potash	Quantity of potash fertilizer applied
Irrigation	Irrigation method or frequency used

6. Tools & Technologies

- Python
 - Google Colab
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
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7. Data Pre-Processing and Feature Engineering

The dataset was prepared for analysis using data preprocessing techniques.

Data Cleaning Steps

- Checked data types of all columns
- Identified missing values

- Handled null values
- Removed duplicate records
- Corrected inconsistent data
- Converted data types where required

Feature Engineering

New features were created to improve analysis.

Examples:

- Age groups were created from the Age column
- Purchase categories were created based on Total Spend
- Additional calculated fields were generated where required

8. Analysis and Visualizations

The cleaned dataset was analysed using **Python** to understand the characteristics of paddy cultivation data. **Exploratory Data Analysis (EDA)** and statistical analysis were performed to identify patterns, trends, and relationships among the variables. Various visualizations, including histograms, box plots, count plots, scatter plots, heatmaps, pair plots, and violin plots, were created using **Matplotlib** and **Seaborn** to explore the data and present meaningful insights into the factors affecting paddy cultivation and yield.

9. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the structure and characteristics of the paddy dataset.

Initial inspection included:

- Data frame structure verification
- Statistical summary of numerical columns
- Missing value verification
- Duplicate record validation
- Data type verification
- Distribution analysis of numerical and categorical variables

EDA helped in understanding the distribution, patterns, and characteristics of the data, providing a strong foundation for further analysis.

10. Statistical Analysis

Statistical analysis was performed to summarize and understand the numerical characteristics of the paddy dataset.

Statistical measures included:

- Mean
- Median
- Mode
- Minimum value
- Maximum value
- Standard deviation
- Variance

These measures helped to understand the central tendency, variability, and overall distribution of the data, providing valuable insights into the characteristics of the paddy cultivation variables.

11. Data Visualization

Data visualization techniques were used to represent the paddy dataset graphically and make patterns, trends, and relationships easier to understand. Various charts created using **Matplotlib** and **Seaborn** helped analyse the distribution of variables, compare categories, identify outliers, and examine relationships among agricultural factors affecting paddy cultivation.

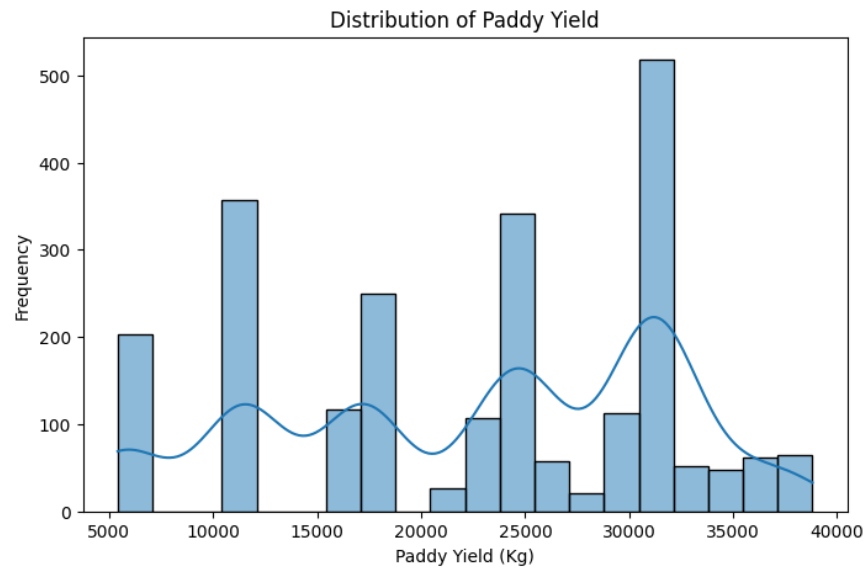
1. Univariate Analysis

Univariate analysis was performed to examine the distribution of individual variables in the paddy dataset.

Analysis Performed:

1. Histogram – Distribution of Paddy Yield

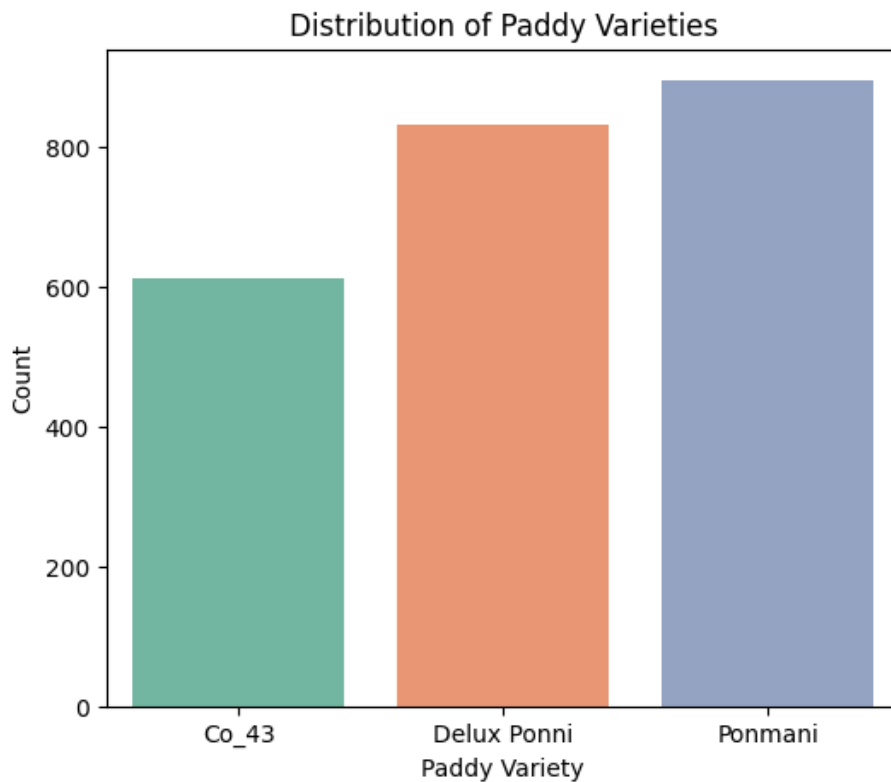
Interpretation: The histogram shows the distribution of paddy yield values, helping to understand the spread, frequency, and overall pattern of crop production



2.

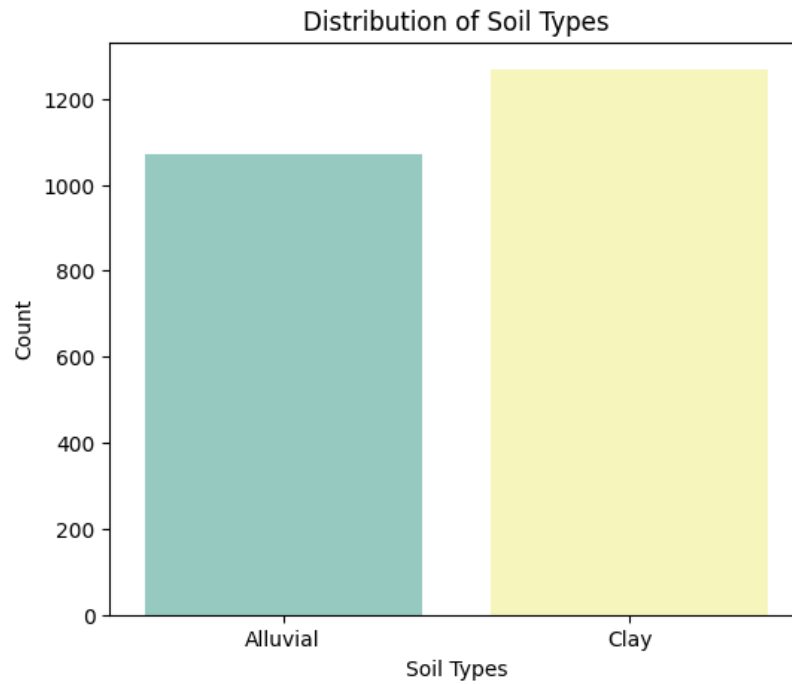
2. Count Plot – Paddy Variety

Interpretation: The count plot compares the frequency of different paddy varieties and identifies the most commonly cultivated variety in the dataset.



3. Count Plot – Soil Types

Interpretation: The count plot shows the distribution of soil types used for paddy cultivation, highlighting the most prevalent soil type in the dataset.



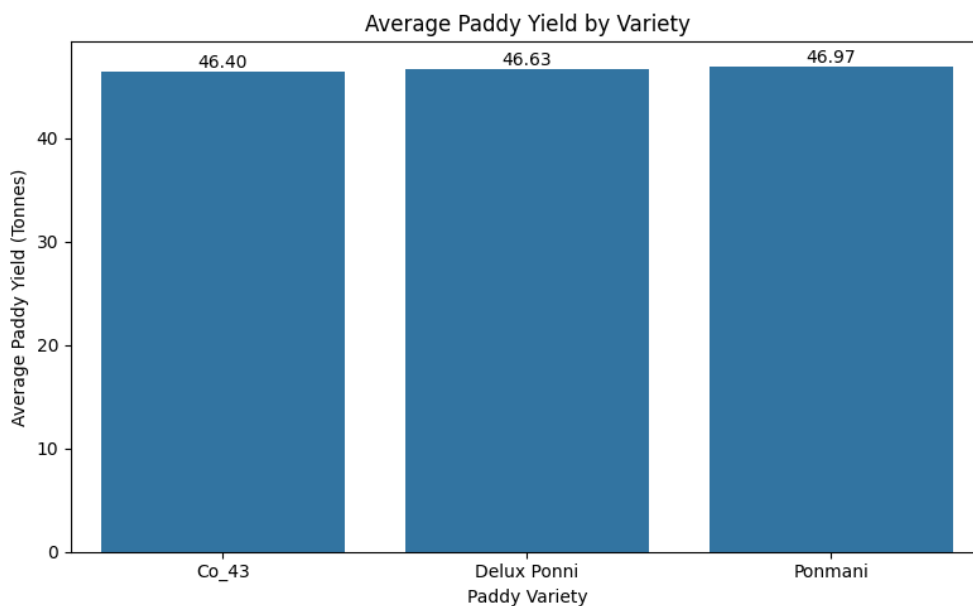
2. Bivariate Analysis

Bivariate analysis was performed to examine the relationship between two variables and understand how different factors influence paddy yield.

Analysis Performed:

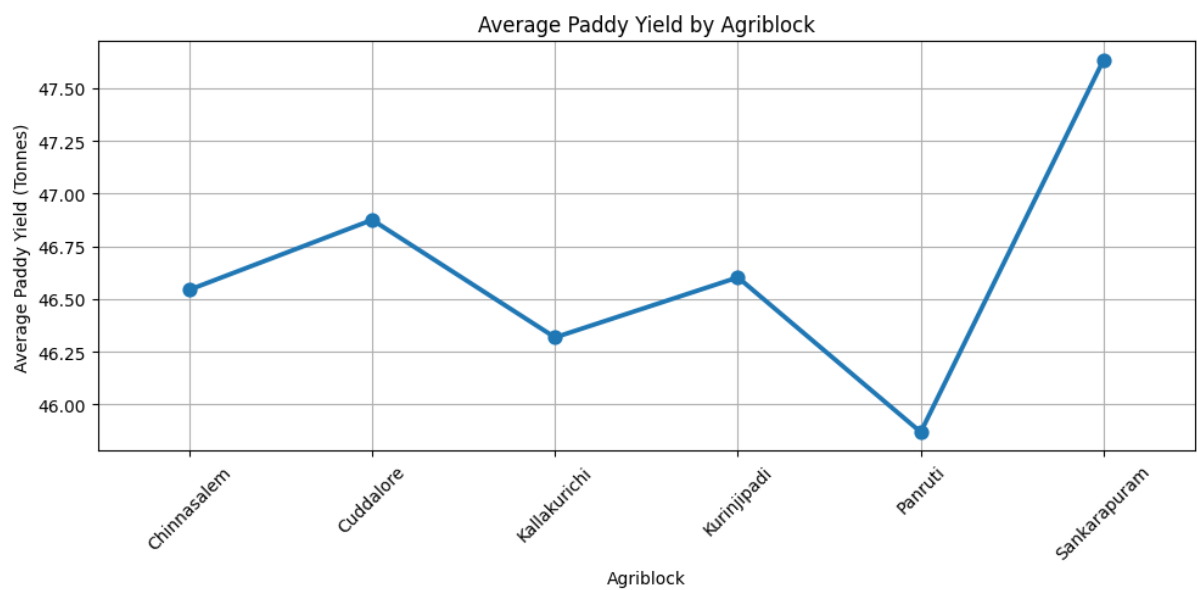
1. Bar Plot (Average Paddy Yield by Variety)

Interpretation: The bar plot compares the average paddy yield of each paddy variety in the dataset. Each bar represents a paddy variety (Ponmani, Delux Ponmani, and CO-43), and the height of the bar indicates its mean paddy yield in tonnes. This visualization helps compare the overall productivity of the different varieties.



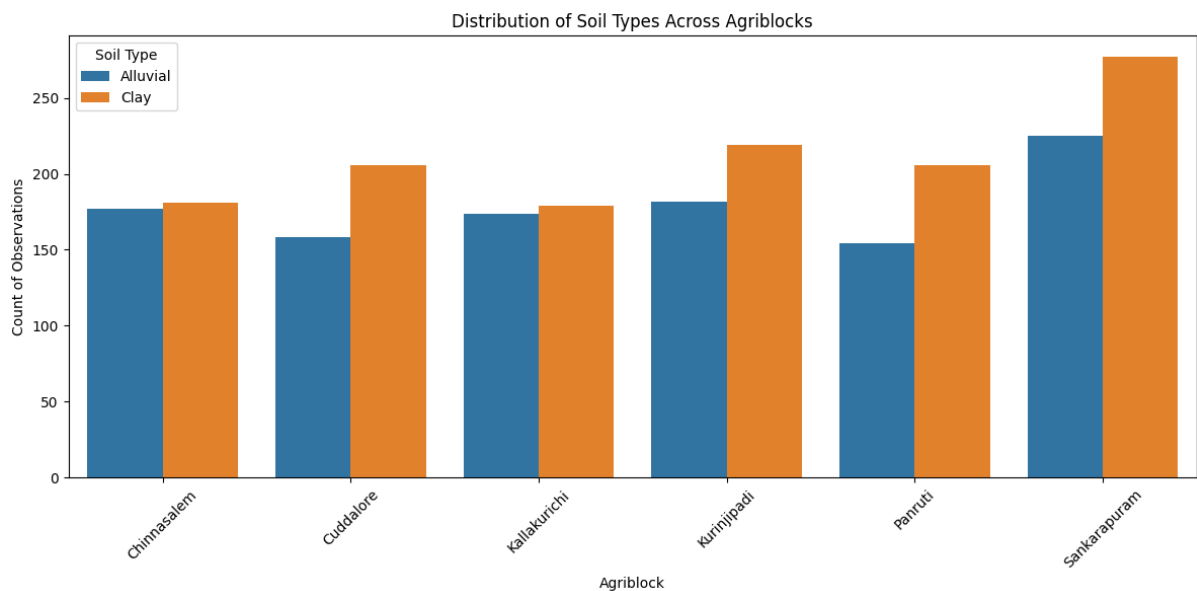
2. Line Plot – Agriblock vs Paddy Yield

Interpretation: The line plot illustrates the average paddy yield across different agriblocks. Each point represents the mean paddy yield of an agriblock, and the connecting line helps visualize how the average yield changes from one agriblock to another.



3. Count Plot with Hue – Agriblock vs Soil Types

Interpretation: the count plot compares the number of **Alluvial** and **Clay** soil observations across different agriblocks. Each agriblock is represented on the x-axis, while different colored bars indicate the count of each soil type within that agriblock.



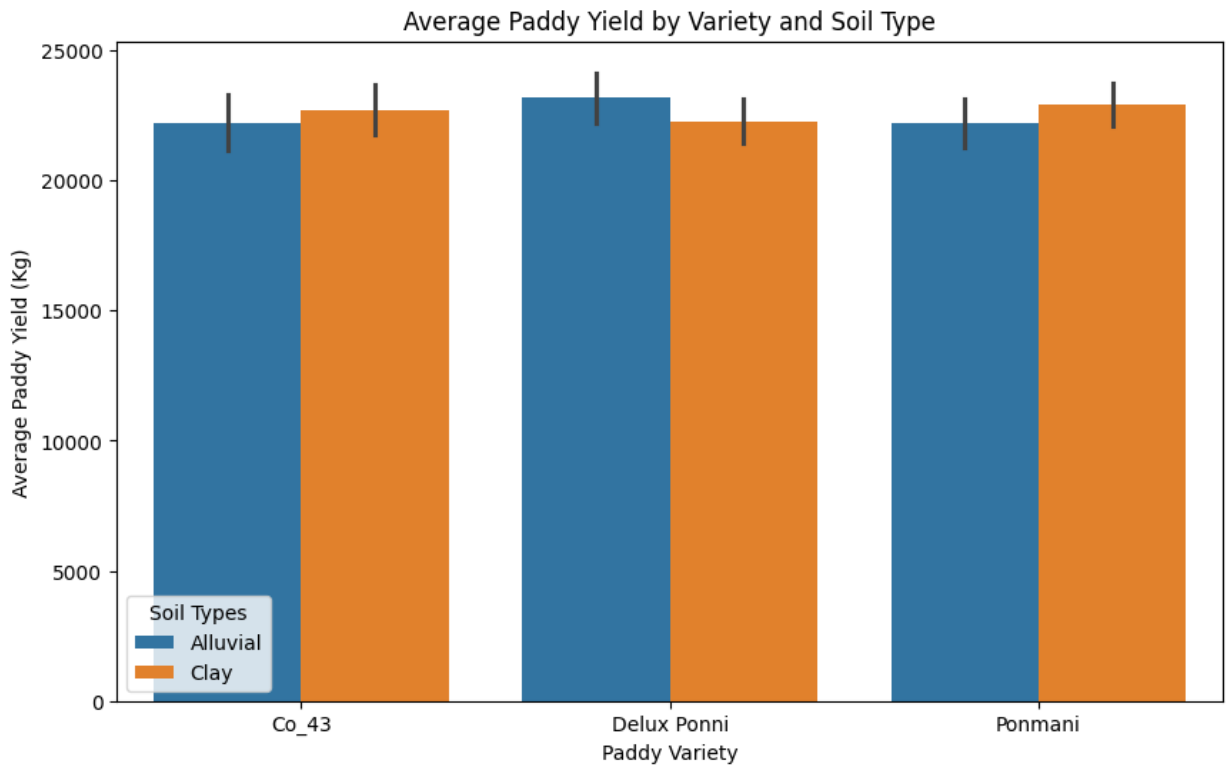
3. Multivariate Analysis

Multivariate analysis was performed to examine the relationship among multiple variables and understand their combined effect on paddy yield.

Analysis Performed:

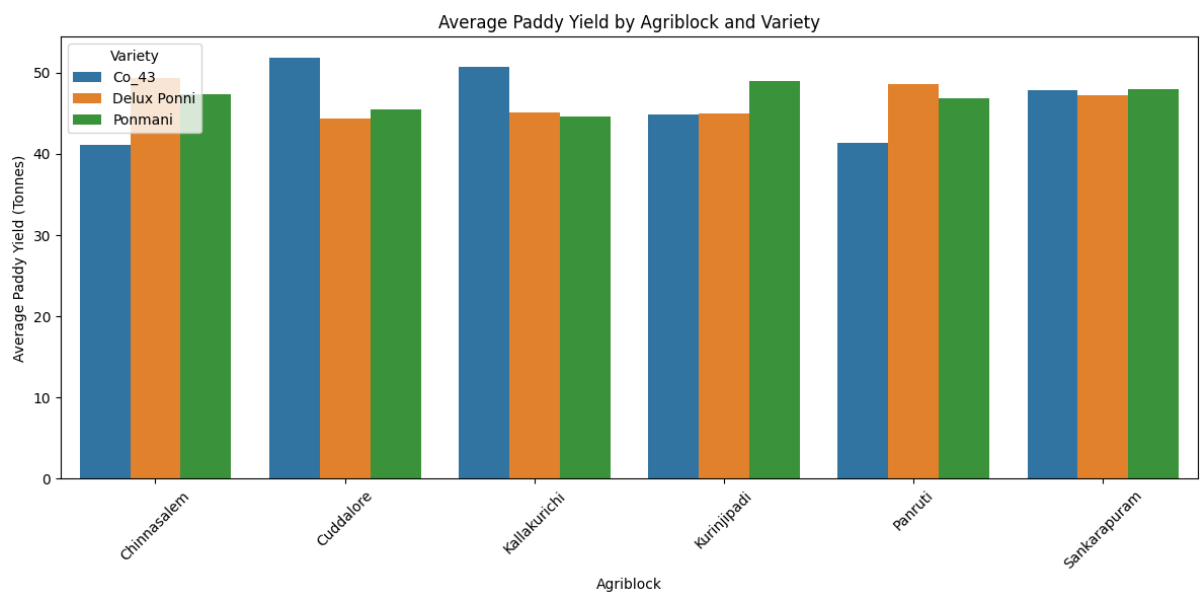
1. Grouped Bar Chart – Variety × Soil Type vs Paddy Yield

Interpretation: The grouped bar chart compares the average paddy yield of three paddy varieties (CO-43, Delux Ponni, and Ponmani) under two different soil types (Alluvial and Clay). Each pair of bars represents the average yield of a variety grown in each soil type, allowing comparison of how soil type influences the productivity of each variety.



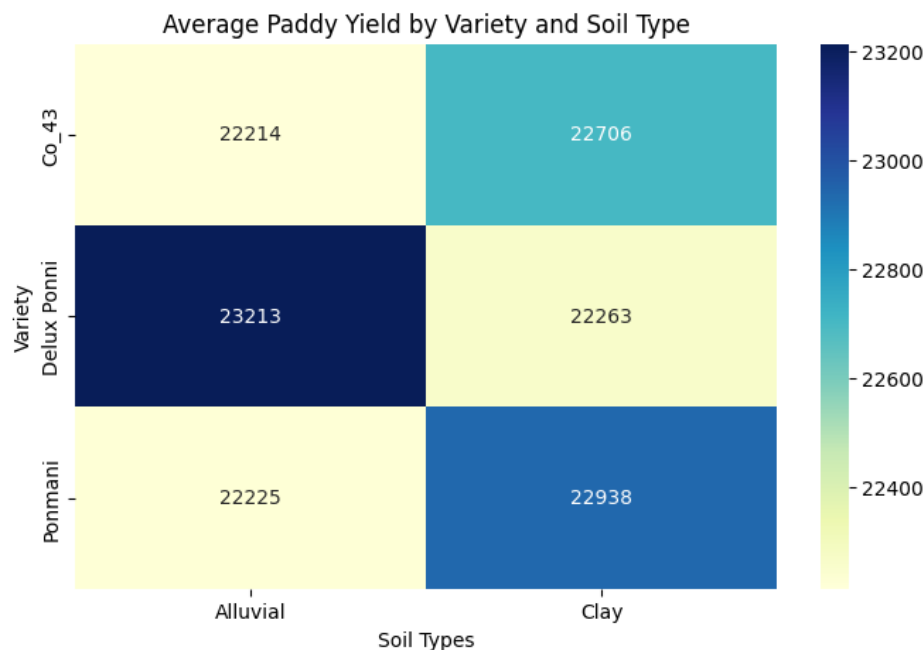
2. Grouped Bar Plot - Agriblock + Variety + Paddy Yield

Interpretation: The grouped bar plot compares the average paddy yield of different paddy varieties across each agriblock. Each agriblock contains separate bars for CO-43, Delux Ponni, and Ponmani, making it easy to compare the performance of varieties within and across locations.



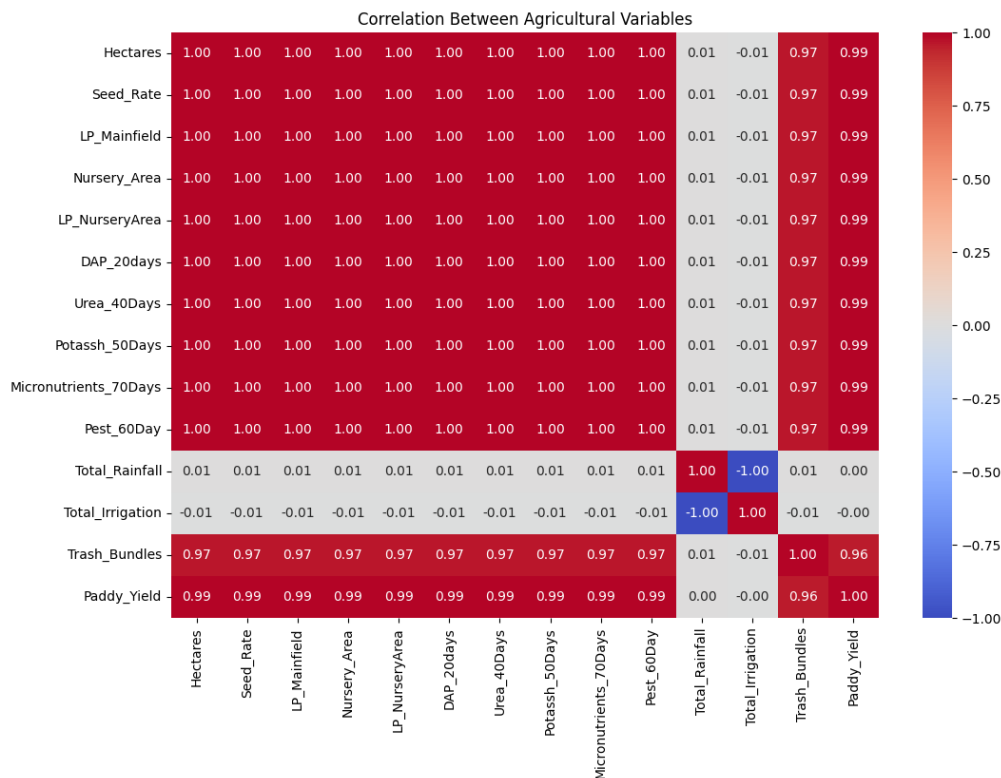
3. Pivot Table Heatmap

Interpretation: The pivot table heatmap displays the average paddy yield across combinations of key categorical variables, making it easier to identify high- and low-performing groups.



4. Correlation Heatmap (with image)

Interpretation: The correlation heatmap illustrates the strength and direction of relationships among numerical variables, helping identify factors that are positively or negatively associated with paddy yield.



12. Summary of Findings

- The analysis of the Paddy Crop dataset revealed that agricultural factors such as **land area, seed rate, fertilizer application, nursery practices, and crop management activities** have a significant influence on paddy yield.
 - The dataset showed variations in productivity across different **paddy varieties and soil types**, indicating that crop selection and soil conditions play an important role in achieving higher yields.
 - Statistical analysis and visualizations identified strong relationships between **cultivation practices (DAP, Urea, Potash, Micronutrients, Pest Control, and LP_Mainfield)** and paddy yield, suggesting that proper resource management contributes to improved productivity.
 - Different **Agriblocks and farming practices** exhibited variations in yield performance, highlighting the impact of regional cultivation patterns and agricultural methods.
 - Correlation analysis indicated that most agricultural inputs had a **positive association with paddy yield**, while some environmental factors showed weaker relationships.
 - Data exploration helped identify important factors affecting crop production and provided insights that can support **better crop planning, resource optimization, and productivity improvement in paddy farming**.
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13. Types of Analysis

1. Descriptive Analysis

Descriptive analysis summarizes and describes the characteristics of the paddy dataset.

Examples:

- Distribution of paddy yield
 - Count of paddy varieties
 - Distribution of soil types
 - Summary statistics of cultivation variables
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2. Diagnostic Analysis

Diagnostic analysis helps identify the factors influencing paddy yield.

Examples:

- Impact of paddy variety on yield
 - Effect of soil type on crop production
 - Relationship between rainfall and paddy yield
 - Effect of fertilizer application on paddy yield
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3. Predictive Analysis

Predictive analysis estimates possible outcomes based on historical data.

Examples:

- Identifying cultivation factors associated with higher paddy yield
 - Analysing relationships among agricultural variables to understand yield patterns
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4. Prescriptive Analysis

Prescriptive analysis provides recommendations based on the insights obtained from the data.

Examples:

- Selecting suitable paddy varieties for improved yield
 - Adopting appropriate soil and fertilizer management practices
 - Optimizing cultivation practices based on rainfall conditions
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14. Future Enhancements and Suggestions

- Include additional paddy cultivation data from different regions and seasons for more comprehensive analysis.
 - Perform advanced statistical analysis to gain deeper insights into crop production factors.
 - Develop interactive dashboards using Power BI or Tableau for better data visualization.
 - Automate the data cleaning, analysis, and reporting process using Python.
 - Incorporate real-time weather and agricultural data for more detailed analysis.
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15. Conclusion

The Paddy Crop Analysis showed that paddy yield varies based on **variety, soil type, and Agriblock (Locations) conditions**. Among the analyzed varieties (**Delux Ponni, Ponmani, and CO-43**), better-performing varieties showed higher productivity under suitable cultivation conditions. Agriblock-wise analysis identified variations in production due to differences in farming practices and resource utilization. Overall, selecting suitable paddy varieties, maintaining proper soil conditions, and improving cultivation practices are essential for achieving higher paddy yield.